

NVIDIA NeMo Agent Toolkit - Nemotron NIMs

NemoDesk

Enterprise IT Ticket AI

Agentic IT service-desk automation that triages, routes, and resolves tickets end-to-end.



India Agentic AI Open Hackathon (NVIDIA + gnani.ai) | Track A: Agentic Workflows



github.com/Yash-Kavaiya/nemodesk-it-agent

The Problem



High volume, rising cost

Service desks face relentless ticket inflow and growing headcount cost.



Slow, inconsistent triage

First response lags; categorization and priority vary by agent.



Tribal-knowledge fixes

L1 agents hunt through scattered runbooks; resolutions aren't standardized.



Sensitive data exposure

Tickets routinely contain emails, IPs, and credentials that leak into tools.

The Opportunity



60-80%

of L1 tickets are repetitive & runbook-resolvable



1st touch

agentic AI can own triage -> route -> resolve -> record



24/7

consistent, instant first response with no fatigue

NemoDesk targets measurable auto-resolution, faster MTTR, and higher SLA adherence.

Solution: NemoDesk

A team of cooperating agents takes a raw ticket and returns a complete, auditable outcome:



Category



Priority P1-P4



Owing team + SLA



Runbook steps

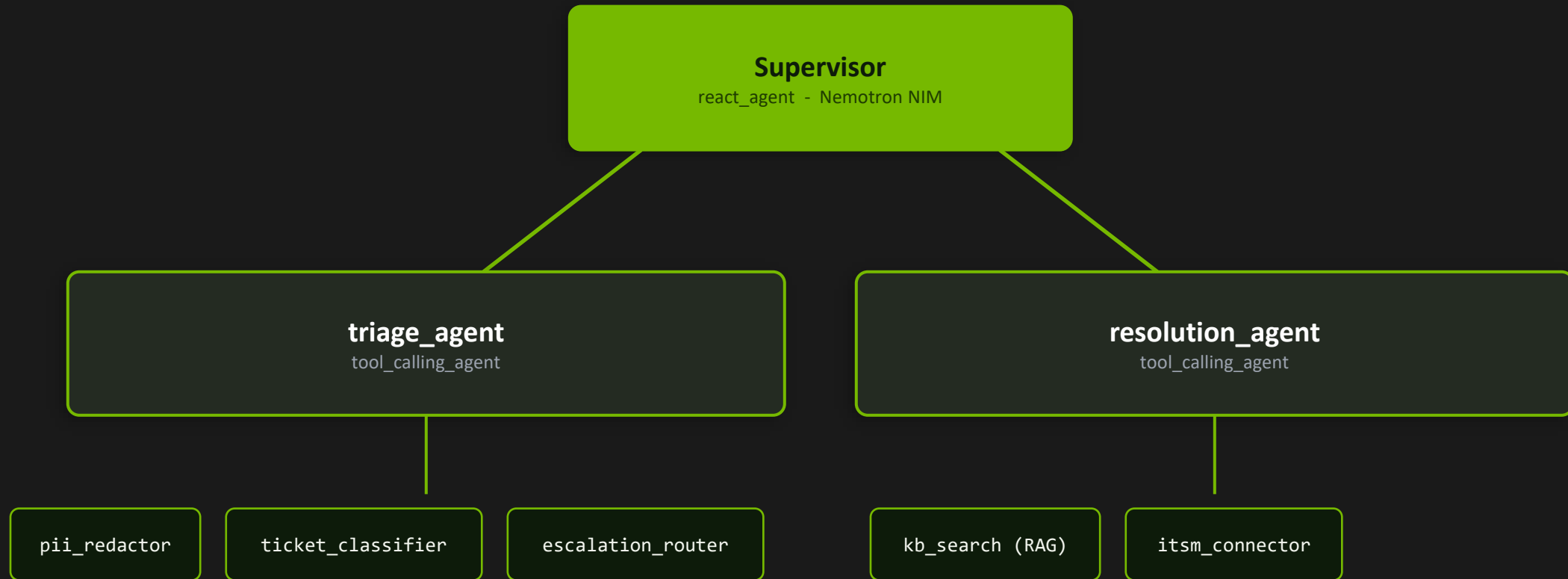


ITSM ticket id



PII redacted first

Architecture



Canonical NeMo Agent Toolkit multi-agent pattern: a reasoning supervisor delegates to focused expert agents, each with a small tool surface.

Built on the NVIDIA Stack



NeMo Agent Toolkit

Multi-agent orchestration, evaluation, observability; MCP & A2A ready.



Nemotron NIM

nemotron-3-nano-30b-a3b for supervisor reasoning + deterministic tool calls.



nv-embedqa-e5-v5

High-quality embeddings powering runbook RAG retrieval.



NIM-portable

Run on build.nvidia.com or on-prem NIM containers - no tool code changes.

Live Demo

P1 outage

"VPN is down for the whole office"

-> Network-Operations, SLA 15 min, escalated

Phishing report

"I entered my password on a fake page"

-> PII redacted, escalated to SecOps-CSIRT as P1

Outlook crash

"Outlook crashes after the latest update"

-> RAG pulls RB-001, numbered fix, INC ticket created

Enterprise-Grade by Design



PII redaction at the boundary

Deterministic masking of emails, IPs, phones, cards, secrets before any LLM call.



Grounded resolutions

RAG over approved runbooks - no hallucinated fixes.



Deterministic SLA & routing

Explicit priority->team and SLA tables: testable and predictable.



Auditable system of record

Every ticket persisted with a unique INC id, category, priority, team.



Graceful degradation

Keyword + severity fallback if a model call cannot be parsed.



Framework-agnostic

NAT instrumentation, profiling and eval out of the box.

Evaluation

nat eval with RAGAS metrics, judged by a Nemotron NIM.



AnswerAccuracy

Is the triage + resolution correct vs. the labelled answer?



ResponseGroundedness

Are the steps grounded in retrieved runbooks, not invented?



10-ticket dataset

Labelled across all 8 categories, incl. P1 security cases.

Run: `nat eval --config_file configs/eval_config.yml`

Impact & ROI

Deflected tickets x avg handle time x loaded agent cost = \$ saved



Faster MTTR



Fewer SLA breaches



Consistent quality



24/7 coverage

Roadmap

1

MCP server

Expose NemoDesk tools for other agents to consume.

2

A2A mesh

Register NemoDesk as a distributed agent node.

3

Real ITSM connectors

Swap mock store for ServiceNow / Jira REST.

4

RL fine-tuning

Train the classifier on historical resolved tickets with NeMo.

5

Observability

Native Phoenix / LangSmith tracing in NAT config.



Let's deploy NemoDesk

Enterprise IT ticket automation, agentic and grounded - built on NVIDIA NeMo Agent Toolkit.

What we need: compute, mentorship, and pilot ticket data.



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